Three Phase Closed Loop Control of MMCs with PR controllers

Sounak Nandi

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1 Introduction

The algorithm for the closed loop simulation of a 3-ϕ MMC is implemented using a $.m$ file in MATLAB 2020b.

1.1 Calculation of the steady state Parameters

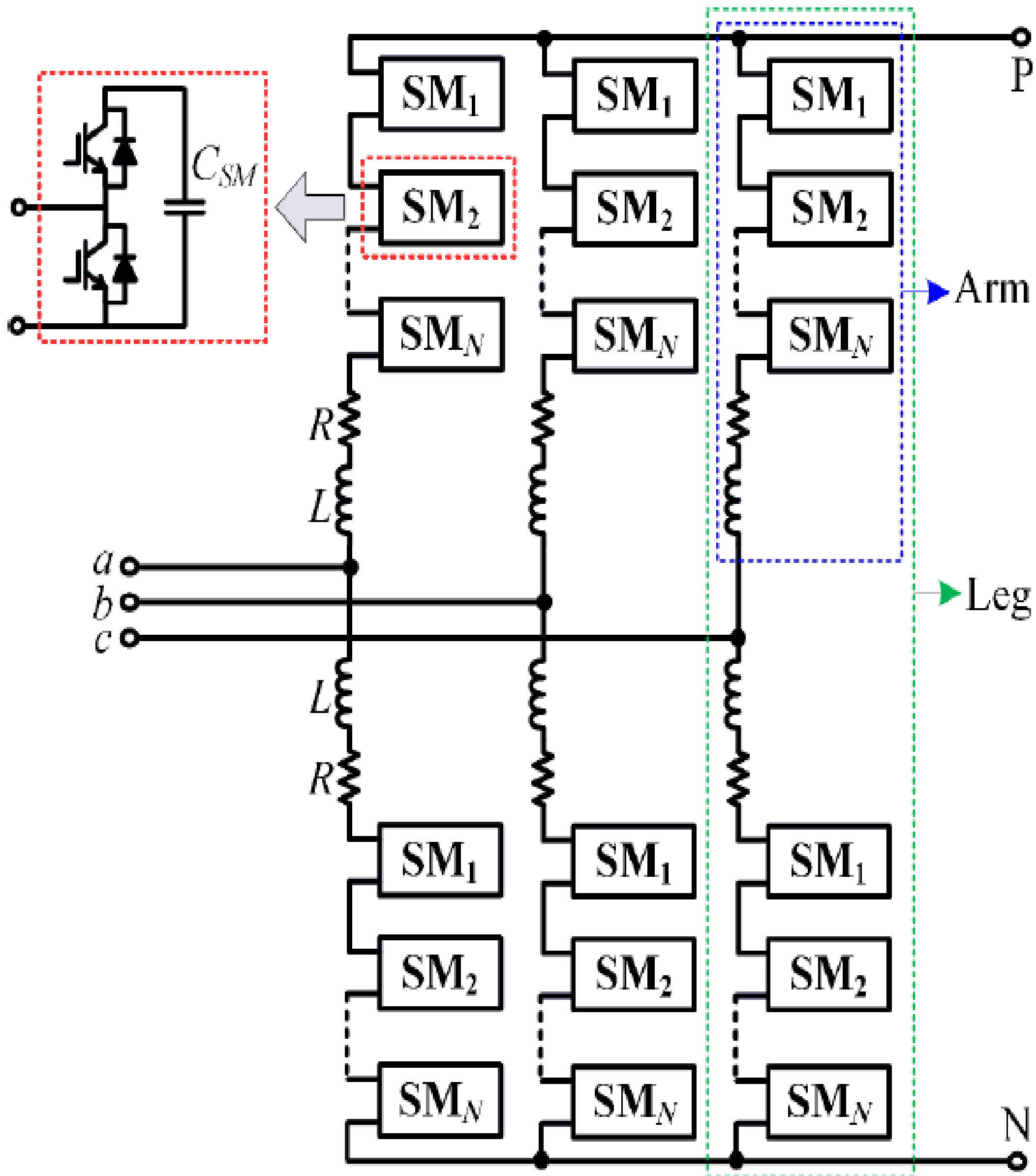

Figure 1: Typical Circuit Diagram of MMC[From 3]

The state space equations are derived based on the following Figures 1and 2.

The state equations are:

$$\frac{di_{cx}}{dt} = -\frac{R}{L}i_{cx} - \frac{n_{ux}}{2L}v^{\Sigma}_{cux} - \frac{n_{lx}}{2L}v^{\Sigma}_{clx} + \frac{V_{dc}}{2L}[x = a, b, c] \tag{1}$$

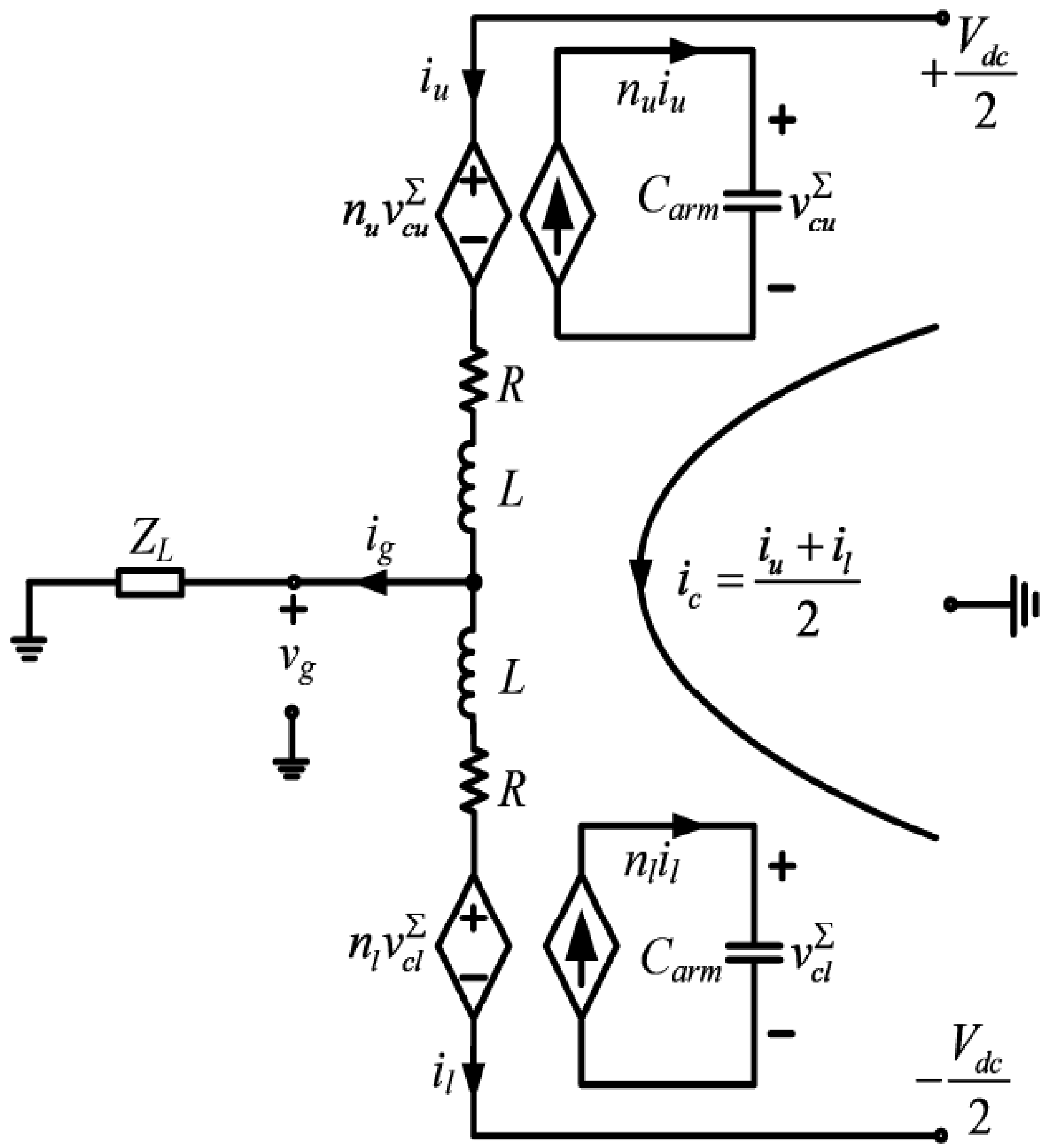

Figure 2: Averaged Value Model of One-Phase of a MMC[From 3]

$$\frac{dv_{cux}^{\Sigma}}{dt} = \frac{n_{ux}}{C_{arm}} i_{cx} - \frac{n_{ux}}{2C_{arm}} i_{gx} [x = a, b, c] \tag{2}$$

$$\frac{dv_{clx}^{\Sigma}}{dt} = \frac{n_{lx}}{C_{arm}} i_{cx} - \frac{n_{lx}}{2C_{arm}} i_{gx} [x = a, b, c] \tag{3}$$

$$\frac{di_{gx}}{dt} = -\frac{n_{ux}}{L} v_{cux}^{\Sigma} + \frac{n_{lx}}{L} v_{clx}^{\Sigma} - \frac{R}{L} i_{gx} - 2\frac{v_{gx}}{L} [x = a, b, c] \tag{4}$$

$$v_{gx} = Z_L i_{gx} [x = a, b, c] \tag{5}$$

$$x(t) = [i_{ca}, i_{cb}, i_{cc}, v_{cua}^{\Sigma}, v_{cub}^{\Sigma}, v_{cuc}^{\Sigma}, v_{cla}^{\Sigma}, v_{clb}^{\Sigma}, v_{clc}^{\Sigma}, i_{ga}, i_{gb}, i_{gc}]^T \tag{6}$$

$$u(t) = [v_{dc}]^T \tag{7}$$

where, i_{cx}=circulating current in the phases[x=a,b,c]
v_{cux}=capacitor voltages of the upper arm[x=a,b,c]
v_{clx}=capacitor voltages of the lower arm[x=a,b,c]
i_{gx}=AC side phase currents[x=a,b,c]
$n_{ux} = 0.5(1 - m)$=switching function of the upper capacitors[x=a,b,c]
$n_{lx} = 0.5(1 + m)$=switching function of the lower capacitors[x=a,b,c]
Now,$n_{ua} = 0.5(1 - m * cos(\omega t))$,$n_{ub} = 0.5(1 - m * cos(\omega t - \frac{2\pi}{3}))$ and $n_{uc} = 0.5(1 - m * cos(\omega t + \frac{2\pi}{3}))$.
and, $n_{la} = 0.5(1 + m * cos(\omega t))$,$n_{lb} = 0.5(1 + m * cos(\omega t - \frac{2\pi}{3}))$ and $n_{lc} = 0.5(1 + m * cos(\omega t + \frac{2\pi}{3}))$.

The $A(t)$ matrix is:

$$A(t) = \begin{pmatrix} -\frac{R}{L} & -\frac{n_{ux}}{2L} & -\frac{n_{lx}}{2L} & 0 \\ \frac{n_{ux}}{C_{arm}} & 0 & 0 & \frac{n_{ux}}{2C_{arm}} \\ \frac{n_{lx}}{C_{arm}} & 0 & 0 & -\frac{n_{lx}}{2C_{arm}} \\ 0 & -\frac{n_{ux}}{L} & \frac{n_{lx}}{L} & -\frac{R+2Z_L}{L} \end{pmatrix} \quad (8)$$

The matrix $A(t)$ is time dependent because the n_{ux} and n_{ul} are dependent on the time.

$$B(t) = [\frac{1}{2L}, 0, 0, 0]^T \quad (9)$$

In the code:
P=$50 * 10^(6)$ W Rated Power
f=50 Hz; Rated Frequency
w1=2*pi*f;
Vdc=$320 * 10^3$V Rated DC Voltage
Vac=$166 * 10^3$V Rated AC Voltage
N=20; submodule number per arm
Csm=$140 * 10^{-6}$ F submodule capacitor
Carm=Csm/N F;
L=$360 * 10^{-3}$ H arm inductance
R=1 Ω; arm resistance
zl=283.73+(408.84*1i)Ω; load from rated power and assumed power factor of 0.8 lag

1.1.1 Harmonic State Space Modelling

The state equations for a Linear time Periodic System is :

$$sX = (A - N)X + BU \quad (10)$$

$$Y = CX + DU \quad (11)$$

Now A, B, C and D are LTP equivalent matrices of $A(t), B(t), C(t)$ and $D(t)$. The matrix $N = diag(jm\omega_o)$. Getting the Fourier Coefficients of a time-varying signal
The signal $x(t)$ needs to be decomposed into complex Fourier series of the form:

$$x(t) = \sum_{-\infty}^{+\infty} X_n exp(jn\omega_o t) \quad (12)$$

So,$x(t)$ can be written in the form,

$$x(t) = \Gamma(t)X \quad (13)$$

where

$$\Gamma(t) = [exp(-jh\omega_o t, exp(-j(h-1)\omega_o t, ..., 1, ..., exp(j(h-1)\omega_o t, exp(jh\omega_o t] \quad (14)$$

$$X = [X_{-h}(t), X_{-(h-1)}(t), ..., X_0(t), ..., X_{(h-1)}(t), X_h(t)]^T \quad (15)$$

Note: The above series is truncated up to the h harmonic.

Toeplitz Matrix In LTP domain, while performing convolution Toeplitz Transformation is required. The Toeplitz Transformation of a vector of the form

$$SW = [SW_{-h}, ..., SW_0, ..., SW_h] \quad (16)$$

is

$$\Gamma[SW] = \begin{bmatrix} SW_0 & SW_{-1} & SW_{-2} & \cdots \\ SW_1 & SW_0 & SW_{-1} & \cdots \\ SW_2 & SW_1 & SW_0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \quad (17)$$

1.2 Steady state analysis code

In the steady state analysis code the value of m=$\frac{2v_{ll}}{\sqrt{3}v_{dc}}$[modulation index]. The code takes into consideration harmonics up to h=3. So, the size of the Toeplitz transform of each vector will be $(2h+1)*(2h+1)$ matrix. Based on equation 10, we need to apply the Toeplitz transform on the matrix $A(t)$ and $B(t)$. The HSS form of the $A(t)$ matrix will be like, for instance, taking the first row(which is also a matrix). The first row is $[\frac{-R}{L}I - N, O, O, \frac{-\Gamma[N_{ua}]}{2L}, O, O, \frac{-\Gamma[N_{la}]}{2L}, O, O, O, O, O]$. All elements in this block are matrices of order $(2h+1)*(2h+1)$. The operator Γ is used to denote, the Toeplitz matrix of the vector. Based on the expression for n_{ua}, the vector to be used for Toeplitz Transform is ,$[0; 0; -1*(m/4); 0.5; -1*(m/4); 0; 0]$. Similarly for n_{ub} and n_{uc} the vector will be,$[0; 0; (m*(1-(\sqrt(3)*1i)))/8; 0.5; (m*(1+(\sqrt(3)*1i)))/8; 0; 0]$ and $[0; 0; (m*(1+(\sqrt(3)*1i)))/8; 0.5; (m*(1-(\sqrt(3)*1i)))/8; 0; 0]$ respectively.

- The N matrix is of the form $diag[-ih\omega_1, -i(h-1)\omega_1, ..., -i\omega_1, DC, i\omega_1, ..., i(h-1)\omega_1, ih\omega_1]$ of order $(2h+1)*(2H+1)$.
- The I matrix is an Identity matrix of order $(2h+1)*(2h+1)$.
- The O matrix is a null matrix of order $(2h+1)*(2h+1)$.

1.3 Toeplitz form of the switching functions for the upper arm capacitors

$$\Gamma[n_{ua}] = \begin{bmatrix} 0.5 & \frac{-m}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{-m}{4} & 0.5 & \frac{-m}{4} & 0 & 0 & 0 & 0 \\ 0 & \frac{-m}{4} & 0.5 & \frac{-m}{4} & 0 & 0 & 0 \\ 0 & 0 & \frac{-m}{4} & 0.5 & \frac{-m}{4} & 0 & 0 \\ 0 & 0 & 0 & \frac{-m}{4} & 0.5 & \frac{-m}{4} & 0 \\ 0 & 0 & 0 & 0 & \frac{-m}{4} & 0.5 & \frac{-m}{4} \\ 0 & 0 & 0 & 0 & 0 & \frac{-m}{4} & 0.5 \end{bmatrix} \tag{18}$$

$$\Gamma[n_{ub}] = \begin{bmatrix} 0.5 & \frac{m(1-i\sqrt{3})}{8} & 0 & 0 & 0 & 0 & 0 \\ \frac{m(1+i\sqrt{3})}{8} & 0.5 & \frac{m(1-i\sqrt{3})}{8} & 0 & 0 & 0 & 0 \\ 0 & \frac{m(1+i\sqrt{3})}{8} & 0.5 & \frac{m(1-i\sqrt{3})}{8} & 0 & 0 & 0 \\ 0 & 0 & \frac{m(1+i\sqrt{3})}{8} & 0.5 & \frac{m(1-i\sqrt{3})}{8} & 0 & 0 \\ 0 & 0 & 0 & \frac{m(1+i\sqrt{3})}{8} & 0.5 & \frac{m(1-i\sqrt{3})}{8} & 0 \\ 0 & 0 & 0 & 0 & \frac{m(1+i\sqrt{3})}{8} & 0.5 & \frac{m(1-i\sqrt{3})}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{m(1+i\sqrt{3})}{8} & 0.5 \end{bmatrix} \tag{19}$$

$$\Gamma[n_{uc}] = \begin{bmatrix} 0.5 & \frac{m(1+i\sqrt{3})}{8} & 0 & 0 & 0 & 0 & 0 \\ \frac{m(1-i\sqrt{3})}{8} & 0.5 & \frac{m(1+i\sqrt{3})}{8} & 0 & 0 & 0 & 0 \\ 0 & \frac{m(1-i\sqrt{3})}{8} & 0.5 & \frac{m(1+i\sqrt{3})}{8} & 0 & 0 & 0 \\ 0 & 0 & \frac{m(1-i\sqrt{3})}{8} & 0.5 & \frac{m(1+i\sqrt{3})}{8} & 0 & 0 \\ 0 & 0 & 0 & \frac{m(1-i\sqrt{3})}{8} & 0.5 & \frac{m(1+i\sqrt{3})}{8} & 0 \\ 0 & 0 & 0 & 0 & \frac{m(1-i\sqrt{3})}{8} & 0.5 & \frac{m(1+i\sqrt{3})}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{m(1-i\sqrt{3})}{8} & 0.5 \end{bmatrix} \tag{20}$$

1.4 Toeplitz form of the switching functions for the Lower arm capacitors

Based on the expression for n_{la}, the vector to be used for Toeplitz Transform is ,$[0; 0; 1*(m/4); 0.5; 1*(m/4); 0; 0]$. Similarly for n_{lb} and n_{lc} the vector will be,$[0; 0; (m*(-1+(\sqrt{(}3)*1i)))/8; 0.5; -(m*(1+(\sqrt{(}3)*1i)))/8; 0; 0]$ and $[0; 0; -(m*(1+(\sqrt{(}3)*1i)))/8; 0.5; (m*(-1+(\sqrt{(}3)*1i)))/8; 0; 0]$ respectively. Based on the procedure described above, the $\Gamma[n_{la}], \Gamma[n_{lb}]$ and $\Gamma[n_{lc}]$ the Toeplitz conversion can be done.

In Matlab, a function *taut.m* is written to perform the conversion of a vector into Toeplitz form.

1.5 The A and B

$$A = \begin{bmatrix} \frac{-R}{L}I-N & O & O & \frac{-\Gamma[N_{ua}]}{2L} & O & O & \frac{-\Gamma[N_{la}]}{2L} & O & O & O & O & O \\ O & \frac{-R}{L}I-N & O & O & \frac{-\Gamma[N_{ub}]}{2L} & O & O & \frac{-\Gamma[N_{lb}]}{2L} & O & O & O & O \\ O & O & \frac{-R}{L}I-N & O & O & \frac{-\Gamma[N_{uc}]}{2L} & O & O & \frac{-\Gamma[N_{lc}]}{2L} & O & O & O \\ \frac{\Gamma[N_{ua}]}{C_{arm}} & O & O & -N & O & O & O & O & O & \frac{\Gamma[N_{ua}]}{2C_{arm}} & O & O \\ O & \frac{\Gamma[N_{ub}]}{C_{arm}} & O & O & -N & O & O & O & O & O & \frac{\Gamma[N_{ub}]}{2C_{arm}} & O \\ O & O & \frac{\Gamma[N_{uc}]}{C_{arm}} & O & O & -N & O & O & O & O & O & \frac{\Gamma[N_{uc}]}{2C_{arm}} \\ \frac{\Gamma[N_{la}]}{C_{arm}} & O & O & -N & O & O & O & O & O & -\frac{\Gamma[N_{la}]}{2C_{arm}} & O & O \\ O & \frac{\Gamma[N_{lb}]}{C_{arm}} & O & O & -N & O & O & O & O & O & -\frac{\Gamma[N_{lb}]}{2C_{arm}} & O \\ O & O & \frac{\Gamma[N_{lc}]}{C_{arm}} & O & O & -N & O & O & O & O & O & -\frac{\Gamma[N_{lc}]}{2C_{arm}} \\ O & O & O & -\frac{\Gamma N_{ua}}{L} & O & O & \frac{\Gamma N_{la}}{L} & O & O & \frac{-(R+2Z_l}{L}I-N & O & O \\ O & O & O & O & -\frac{\Gamma N_{ub}}{L} & O & O & \frac{\Gamma N_{lb}}{L} & O & O & \frac{-(R+2Z_l}{L}I-N & O \\ O & O & O & O & O & -\frac{\Gamma N_{ub}}{L} & O & O & \frac{\Gamma N_{lb}}{L} & O & O & \frac{-(R+2Z_l}{L}I-N \end{bmatrix} \tag{21}$$

The size of the A matrix is the number of state variables times $(2h+1)$. Here $h=3$, so $2h+1=7$. Therefore the size of a matrix is $12*7=84$. So, the order is $84X84$.

$B = [\frac{1}{2L}I, \frac{1}{2L}I, \frac{1}{2L}I, O, O, O, O, O, O, O, O, O]^T$.

The size of the B matrix is the number of state variables times $(2h+1)$. Here $h=3$, so $2h+1=7$. Therefore the size of a matrix is $12*7=84$.

So, the order is $84X7$.Because the U matrix is of the form of $7X1$ only one input of V_{dc} is used.

1.6 Steady State Response

The steady-state response of the state variables is given by $X_{ss} = -A^{-1}BU$.

Now the state variable response obtained from the equation is only the Fourier Coefficients. To convert them back in the time domain, we need an operator $P(t)$. $x(t) = P(t)X$, where $P(t) = [exp(-ih\omega_1 t), exp(-i(h-1)\omega_1 t), ..., 1, ..., exp(i(h-1)\omega_1 t), exp(ih\omega_1 t)]$ and $X^T = [X_{-h}, X_{-(h-1)}, ..., X_0, ..., X_{h-1}, X_h]$. We obtain the X vector from the steady-state calculations.

In Matlab, the function timedomain.m is written to perform the conversion of a vector into a time domain.

The Output from the steady state analysis is

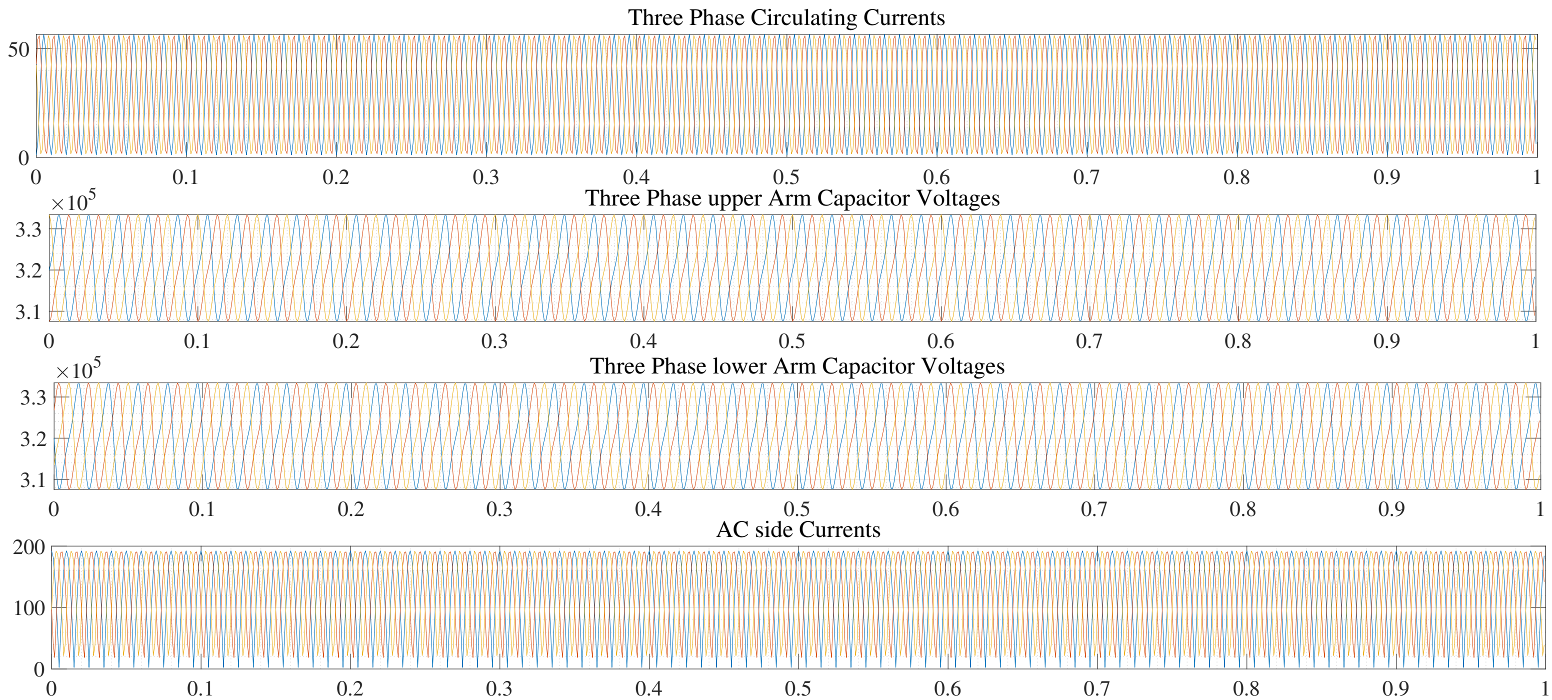

Figure 3: Steady State Output

2 Closed Loop Control Implementation

Linearizing the (1)-(4) equations we get:

$$\frac{d\Delta i_{cx}}{dt} = -\frac{R}{L}\Delta i_{cx} - \frac{n_{ux0}}{2L}\Delta v^{\Sigma}_{cux} - \frac{v^{\Sigma}_{cuxs}}{2L}\Delta n_{ux} - \frac{n_{lx0}}{2L}\Delta v^{\Sigma}_{clx} - \frac{v^{\Sigma}_{clxs}}{2L}\Delta n_{lx} + \frac{1}{2L}\Delta v_{dc}[x = a, b, c] \tag{22}$$

$$\frac{d\Delta v^{\Sigma}_{cux}}{dt} = \frac{n_{ux0}}{C_{arm}}\Delta i_{cx} + \frac{n_{ux0}}{2C_{arm}}\Delta i_{gx} + \frac{1}{C_{arm}}(i_{cxs} + \frac{i_{gxs}}{2})\Delta n_{ux}[x = a, b, c] \tag{23}$$

$$\frac{d\Delta v^{\Sigma}_{clx}}{dt} = \frac{n_{lx0}}{C_{arm}}\Delta i_{cx} - \frac{n_{lx0}}{2C_{arm}}\Delta i_{gx} + \frac{1}{C_{arm}}(i_{cxs} - \frac{i_{gxs}}{2})\Delta n_{lx}[x = a, b, c] \tag{24}$$

$$\frac{d\Delta i_{gx}}{dt} = -\frac{n_{ux0}}{L}\Delta v^{\Sigma}_{cux} - \frac{v^{\Sigma}_{cuxs}}{L}\Delta n_{ux} + \frac{n_{lx0}}{L}\Delta v^{\Sigma}_{clx} + \frac{v^{\Sigma}_{clxs}}{L}\Delta n_{lx} - \frac{R + 2Z_L}{L}\Delta i_{gx}[x = a, b, c] \tag{25}$$

In these above (22)-(25) equations, the subscript 's' and '0' are used in the switching functions and state variables to denote steady-state values, and the symbol Δ is used to denote the small-signal part.

The small-signal controller equations are:

$$\frac{d\Delta x_{prx1}}{dt} = -{\omega_1}^2\Delta x_{prx2} + K_r\Delta v^*_{gx} - K_r Z_L\Delta i_{gx}[x = a, b, c] \tag{26}$$

$$\frac{d\Delta x_{prx2}}{dt} = \Delta x_{prx1}[x = a, b, c] \tag{27}$$

2.1 How x_{prx1} and x_{prx2} are coming?

The transfer function of $H_v(s)$=$K_p + \frac{K_r s}{s^2+\omega_1^2}$. Now, $\Delta x_{prx1}(s) = \frac{K_r s}{s^2+\omega_1^2}(v^*_{gx}(s) - v_{gx}(s))$ and $\Delta x_{prx2}(s) = \frac{K_r(v^*_{gx}(s)-v_{gx}(s))}{s^2+\omega_1^2}$.[x=a,b,c] and $v_{gx}(s) = Z_L(s)i_{gx}(s)$.Please refer 6 and 9 for detailed understanding of the statements.

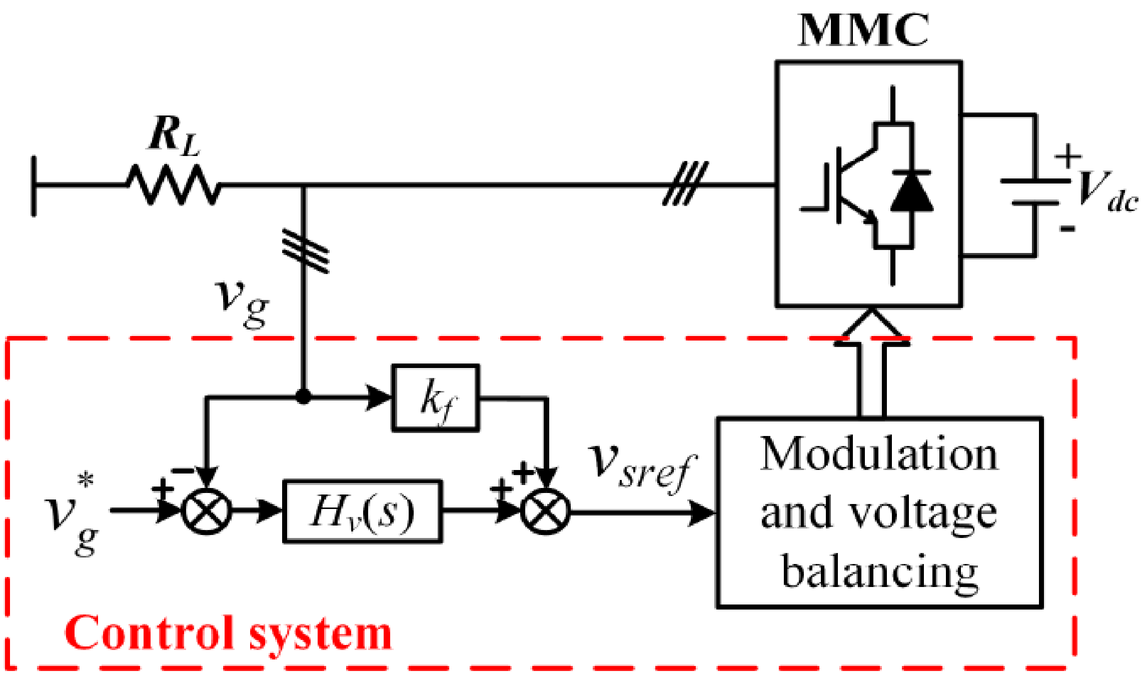

Figure 4: Closed Loop Control using Proportional Resonant Controllers[AC Voltage Control][From 1]

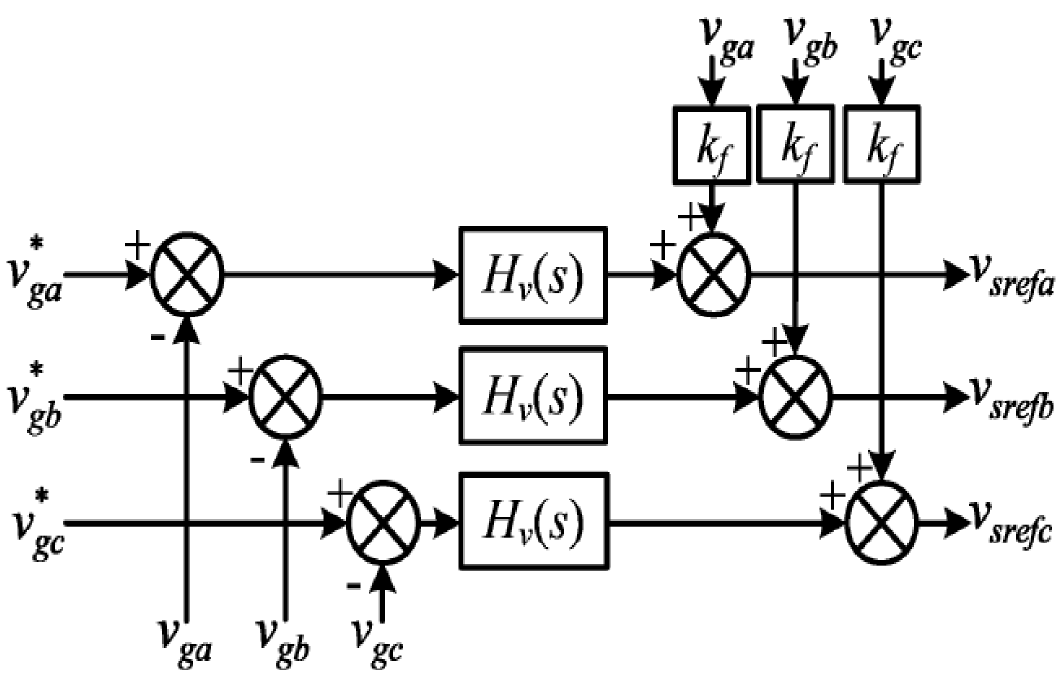

Figure 5: Detailed View of the Controller Portion[From 3]

2.2 Output State equations

The outputs are Δn_{ux} and Δn_{lx} where [x=a,b,c].

$$\Delta n_{ux} = -\frac{1}{V_{dc}}\Delta x_{prx1} - \frac{K_p}{V_{dc}}\Delta v_{gx}^* + \frac{K_p - K_f}{V_{dc}} Z_L \Delta i_{gx}[x = a, b, c] \tag{28}$$

$$\Delta n_{lx} = \frac{1}{V_{dc}}\Delta x_{prx1} + \frac{K_p}{V_{dc}}\Delta v_{gx}^* - \frac{K_p - K_f}{V_{dc}} Z_L \Delta i_{gx}[x = a, b, c] \tag{29}$$

where K_f is the feed-forward gain used in the PR controller. Please refer 6 and 9 for detailed understanding of the statements.

2.3 Small signal HSS and A and B matrices

- The state variables used in this analysis are 18 in number.
- A total of 4 input variables are used.

$$\frac{d\Delta x(t)}{dt} = A(t)\Delta x(t) + B(t)\Delta u(t) \tag{30}$$

The state variables are
$\Delta x(t) = [\Delta i_{ca}, \Delta i_{cb}, \Delta i_{cc}, \Delta v_{cua}^{\sum}, \Delta v_{cub}^{\sum}, \Delta v_{cuc}^{\sum}, \Delta v_{cla}^{\sum}, \Delta v_{clb}^{\sum}, \Delta v_{clc}^{\sum}, \Delta i_{ga}, \Delta i_{gb}, \Delta i_{gc}, \Delta x_{pra1}, \Delta x_{pra2},$
$\Delta x_{prb1}, \Delta x_{prb2}, \Delta x_{prc1}, \Delta x_{prc2},]$.

The input variables are
$\Delta u(t) = [\Delta v_{dc}, \Delta v_{ga}^{*}, \Delta v_{gb}^{*}, \Delta v_{gc}^{*}]$.

$$A(t) = \begin{bmatrix}
\frac{-R}{L} & 0 & 0 & -\frac{N_{ua0}}{2L} & 0 & 0 & -\frac{N_{la0}}{2L} & 0 & 0 & F_{c1a} & 0 & 0 & F_{c2a} & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{-R}{L} & 0 & 0 & -\frac{N_{ub0}}{2L} & 0 & 0 & -\frac{N_{lb0}}{2L} & 0 & 0 & F_{c1b} & 0 & 0 & 0 & F_{c2b} & 0 & 0 & 0 \\
0 & 0 & \frac{-R}{L} & 0 & 0 & -\frac{N_{uc0}}{2L} & 0 & 0 & -\frac{N_{lc0}}{2L} & 0 & 0 & F_{c1c} & 0 & 0 & 0 & 0 & F_{c2c} & 0 \\
\frac{N_{ua0}}{C_{arm}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & F_{vu1a} & 0 & 0 & F_{vu2a} & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{N_{ub0}}{C_{arm}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & F_{vu1b} & 0 & 0 & 0 & F_{vu2b} & 0 & 0 & 0 \\
0 & 0 & \frac{N_{uc0}}{C_{arm}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & F_{vu1c} & 0 & 0 & 0 & 0 & F_{vu2c} & 0 \\
\frac{N_{la0}}{C_{arm}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & F_{vl1a} & 0 & 0 & F_{vl2a} & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{N_{lb0}}{C_{arm}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & F_{vl1b} & 0 & 0 & 0 & F_{vl2b} & 0 & 0 & 0 \\
0 & 0 & \frac{N_{lc0}}{C_{arm}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & F_{vl1c} & 0 & 0 & 0 & 0 & F_{vl2c} & 0 \\
0 & 0 & 0 & -\frac{N_{ua0}}{L} & 0 & 0 & \frac{N_{la0}}{L} & 0 & 0 & F_{i1a} & 0 & 0 & F_{i2a} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{N_{ub0}}{L} & 0 & 0 & \frac{N_{lb0}}{L} & 0 & 0 & F_{i1b} & 0 & 0 & 0 & F_{i2b} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{N_{uc0}}{L} & 0 & 0 & \frac{N_{lc0}}{L} & 0 & 0 & F_{i1c} & 0 & 0 & 0 & 0 & F_{i2c} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -K_r Z_L & 0 & 0 & 0 & -\omega_1^2 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -K_r Z_L & 0 & 0 & 0 & 0 & -\omega_1^2 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -K_r Z_L & 0 & 0 & 0 & 0 & 0 & -\omega_1^2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0
\end{bmatrix} \tag{31}$$

$$B(t) = \begin{bmatrix}
\frac{1}{2L} & F_{c3a} & 0 & 0 \\
\frac{1}{2L} & 0 & F_{c3b} & 0 \\
\frac{1}{2L} & 0 & 0 & F_{c3c} \\
0 & F_{vu3a} & 0 & 0 \\
0 & 0 & F_{vu3b} & 0 \\
0 & 0 & 0 & F_{vu3c} \\
0 & F_{vl3a} & 0 & 0 \\
0 & 0 & F_{vl3b} & 0 \\
0 & 0 & 0 & F_{vl3c} \\
0 & F_{i3a} & 0 & 0 \\
0 & 0 & F_{i3b} & 0 \\
0 & 0 & 0 & F_{i3c} \\
0 & K_r & 0 & 0 \\
0 & 0 & 0 & 0 \\
0 & 0 & K_r & 0 \\
0 & 0 & 0 & 0 \\
0 & 0 & 0 & K_r \\
0 & 0 & 0 & 0
\end{bmatrix} \tag{32}$$

The time-dependent nature of the $A(t)$ and $B(t)$ are coming because of the switching functions and some variables that are defined below.

$$F_{c1x} = \frac{(K_f - K_p)(v_{cuxs}^{\sum} - v_{clxs}^{\sum})}{2LV_{dc}} Z_L ...[x = a, b, c] \tag{33}$$

$$F_{c2x} = \frac{v_{cuxs}^{\sum} - v_{clxs}^{\sum}}{2LV_{dc}} ...[x = a, b, c] \tag{34}$$

$$F_{c3x} = \frac{K_p(v_{cuxs}^{\sum} - v_{clxs}^{\sum})}{2LV_{dc}} ...[x = a, b, c] \tag{35}$$

$$F_{vu1x} = \frac{n_{ux0}}{2C_{arm}} - \frac{(K_f - K_p)}{C_{arm}V_{dc}}(i_{cxs} + \frac{i_{gxs}}{2})Z_L ...[x = a, b, c] \tag{36}$$

$$F_{vu2x} = -\frac{1}{C_{arm}V_{dc}}(i_{cxs} + \frac{i_{gxs}}{2})...[x = a, b, c] \tag{37}$$

$$F_{vu3x} = -\frac{K_p}{C_{arm}V_{dc}}(i_{cxs} + \frac{i_{gxs}}{2})...[x = a, b, c] \tag{38}$$

$$F_{vl1x} = \frac{n_{lx0}}{2C_{arm}} + \frac{(K_f - K_p)}{C_{arm}V_{dc}}(i_{cxs} - \frac{i_{gxs}}{2})Z_L ...[x = a, b, c] \tag{39}$$

$$F_{vl2x} = \frac{1}{C_{arm}V_{dc}}(i_{cxs} - \frac{i_{gxs}}{2})...[x = a, b, c] \tag{40}$$

$$F_{vl3x} = \frac{K_p}{C_{arm}V_{dc}}(i_{cxs} - \frac{i_{gxs}}{2})...[x = a, b, c] \tag{41}$$

$$F_{i1x} = -\frac{R + 2Z_L}{L} + \frac{(K_f - K_p)(v^{\sum}_{cuxs} + v^{\sum}_{clxs})}{LV_{dc}} Z_L ...[x = a, b, c] \tag{42}$$

$$F_{i2x} = \frac{v^{\sum}_{cuxs} + v^{\sum}_{clxs}}{LV_{dc}} ...[x = a, b, c] \tag{43}$$

$$F_{i3x} = \frac{K_p(v^{\sum}_{cuxs} + v^{\sum}_{clxs})}{LV_{dc}} ...[x = a, b, c] \tag{44}$$

2.4 The $C(t)$ and the $D(t)$ matrices

$$C(t) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{K_p-K_f}{V_{dc}}Z_L & 0 & 0 & -\frac{1}{V_{dc}} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{K_p-K_f}{V_{dc}}Z_L & 0 & 0 & -\frac{1}{V_{dc}} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{K_p-K_f}{V_{dc}}Z_L & 0 & 0 & -\frac{1}{V_{dc}} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{K_p-K_f}{V_{dc}}Z_L & 0 & 0 & \frac{1}{V_{dc}} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{K_p-K_f}{V_{dc}}Z_L & 0 & 0 & \frac{1}{V_{dc}} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{K_p-K_f}{V_{dc}}Z_L & 0 & 0 & \frac{1}{V_{dc}} & 0 & 0 & 0
\end{bmatrix} \tag{45}$$

$$D(t) = \begin{bmatrix}
0 & -\frac{K_p}{V_{dc}} & 0 & 0 \\
0 & 0 & -\frac{K_p}{V_{dc}} & 0 \\
0 & 0 & 0 & -\frac{K_p}{V_{dc}} \\
0 & \frac{K_p}{V_{dc}} & 0 & 0 \\
0 & 0 & \frac{K_p}{V_{dc}} & 0 \\
0 & 0 & 0 & \frac{K_p}{V_{dc}}
\end{bmatrix} \tag{46}$$

$$\Delta y(t) = [\Delta n_{ua}, \Delta n_{ub}, \Delta n_{uc}, \Delta n_{la}, \Delta n_{lb}, \Delta n_{lc}]^T \tag{47}$$

2.5 HSS-based transformation of the matrices

$$A = \begin{bmatrix}
\frac{-R}{L}-N & O & O & -\frac{\Gamma[N_{ua0}]}{2L} & O & O & -\frac{\Gamma[N_{la0}]}{2L} & O & O & \Gamma[F_{c1a}] & O & O & \Gamma[F_{c2a}] & O & O & O & O & O \\
O & \frac{-R}{L}-N & O & O & -\frac{\Gamma[N_{ub0}]}{2L} & O & O & -\frac{\Gamma[N_{lb0}]}{2L} & O & O & \Gamma[F_{c1b}] & O & O & O & \Gamma[F_{c2b}] & O & O & O \\
O & O & \frac{-R}{L}-N & O & O & -\frac{\Gamma[N_{uc0}]}{2L} & O & O & -\frac{\Gamma[N_{lc0}]}{2L} & O & O & \Gamma[F_{c1c}] & O & O & O & O & \Gamma[F_{c2c}] & O \\
\frac{\Gamma[N_{ua0}]}{C_{arm}} & O & O & -N & O & O & O & O & O & \Gamma[F_{vu1a}] & O & O & \Gamma[F_{vu2a}] & O & O & O & O & O \\
O & \frac{\Gamma[N_{ub0}]}{C_{arm}} & O & O & -N & O & O & O & O & O & \Gamma[F_{vu1b}] & O & O & O & \Gamma[F_{vu2b}] & O & O & O \\
O & O & \frac{\Gamma[N_{uc0}]}{C_{arm}} & O & O & -N & O & O & O & O & O & \Gamma[F_{vu1c}] & O & O & O & O & \Gamma[F_{vu2c}] & O \\
\frac{\Gamma[N_{la0}]}{C_{arm}} & O & O & O & O & O & -N & O & O & \Gamma[F_{vl1a}] & O & O & \Gamma[F_{vl2a}] & O & O & O & O & O \\
O & \frac{\Gamma[N_{lb0}]}{C_{arm}} & O & O & O & O & O & -N & O & O & \Gamma[F_{vl1b}] & O & O & O & \Gamma[F_{vl2b}] & O & O & O \\
O & O & \frac{\Gamma[N_{lc0}]}{C_{arm}} & O & O & O & O & O & -N & O & O & \Gamma[F_{vl1c}] & O & O & O & O & \Gamma[F_{vl2c}] & O \\
O & O & O & -\frac{\Gamma[N_{ua0}]}{L} & O & O & \frac{\Gamma[N_{la0}]}{L} & O & O & \Gamma[F_{i1a}]-N & O & O & \Gamma[F_{i2a}] & O & O & O & O & O \\
O & O & O & O & -\frac{\Gamma[N_{ub0}]}{L} & O & O & \frac{\Gamma[N_{lb0}]}{L} & O & O & \Gamma[F_{i1b}]-N & O & O & O & \Gamma[F_{i2b}] & O & O & O \\
O & O & O & O & O & -\frac{\Gamma[N_{uc0}]}{L} & O & O & \frac{\Gamma[N_{lc0}]}{L} & O & O & \Gamma[F_{i1c}]-N & O & O & O & O & \Gamma[F_{i2c}] & O \\
O & O & O & O & O & O & O & O & O & -K_r Z_L I & O & O & -N & -\omega_1^2 I & O & O & O & O \\
O & O & O & O & O & O & O & O & O & O & O & O & I & -N & O & O & O & O \\
O & O & O & O & O & O & O & O & O & O & -K_r Z_L I & O & O & O & -N & -\omega_1^2 I & O & O \\
O & O & O & O & O & O & O & O & O & O & O & O & O & O & I & -N & O & O \\
O & O & O & O & O & O & O & O & O & O & O & -K_r Z_L I & O & O & O & O & -N & -\omega_1^2 I \\
O & O & O & O & O & O & O & O & O & O & O & O & O & O & O & O & I & -N
\end{bmatrix} \tag{48}$$

$$B = \begin{bmatrix} \frac{1}{2L}I & \Gamma[F_{c3a}] & O & O \\ \frac{1}{2L}I & O & \Gamma[F_{c3b}] & O \\ \frac{1}{2L}I & O & O & \Gamma[F_{c3c}] \\ O & \Gamma[F_{vu3a}] & O & O \\ O & O & \Gamma[F_{vu3b}] & O \\ O & O & O & \Gamma[F_{vu3c}] \\ O & \Gamma[F_{vl3a}] & O & O \\ O & O & \Gamma[F_{vl3b}] & O \\ O & O & O & \Gamma[F_{vl3c}] \\ O & \Gamma[F_{i3a}] & O & O \\ O & O & \Gamma[F_{i3b}] & O \\ O & O & O & \Gamma[F_{i3c}] \\ O & K_r I & O & O \\ O & O & O & O \\ O & O & K_r I & O \\ O & O & O & O \\ O & O & O & K_r I \\ O & O & O & O \end{bmatrix} \tag{49}$$

$$C = \begin{bmatrix} O & O & O & O & O & O & O & O & O & \frac{K_p - K_f}{V_{dc}} Z_L I & O & O & -\frac{1}{V_{dc}} I & O & O & O & O & O \\ O & O & O & O & O & O & O & O & O & O & \frac{K_p - K_f}{V_{dc}} Z_L I & O & O & -\frac{1}{V_{dc}} I & O & O & O & O \\ O & O & O & O & O & O & O & O & O & O & O & \frac{K_p - K_f}{V_{dc}} Z_L I & O & O & -\frac{1}{V_{dc}} I & O & O & O \\ O & O & O & O & O & O & O & O & O & -\frac{K_p - K_f}{V_{dc}} Z_L I & O & O & \frac{1}{V_{dc}} I & O & O & O & O & O \\ O & O & O & O & O & O & O & O & O & O & -\frac{K_p - K_f}{V_{dc}} Z_L I & O & O & \frac{1}{V_{dc}} I & O & O & O & O \\ O & O & O & O & O & O & O & O & O & O & O & -\frac{K_p - K_f}{V_{dc}} Z_L I & O & O & \frac{1}{V_{dc}} I & O & O & O \end{bmatrix} \tag{50}$$

$$D = \begin{bmatrix} O & -\frac{K_p}{V_{dc}} I & O & O \\ O & O & -\frac{K_p}{V_{dc}} I & O \\ O & O & O & -\frac{K_p}{V_{dc}} I \\ O & \frac{K_p}{V_{dc}} I & O & O \\ O & O & \frac{K_p}{V_{dc}} I & o \\ O & O & O & \frac{K_p}{V_{dc}} I \end{bmatrix} \tag{51}$$

3 References

1 J. Lyu, X. Zhang, X. Cai and M. Molinas, "Harmonic State-Space Based Small-Signal Impedance Modeling of a Modular Multilevel Converter With Consideration of Internal Harmonic Dynamics," in IEEE Transactions on Power Electronics, vol. 34, no. 3, pp. 2134-2148, March 2019, doi: 10.1109/TPEL.2018.2842682.

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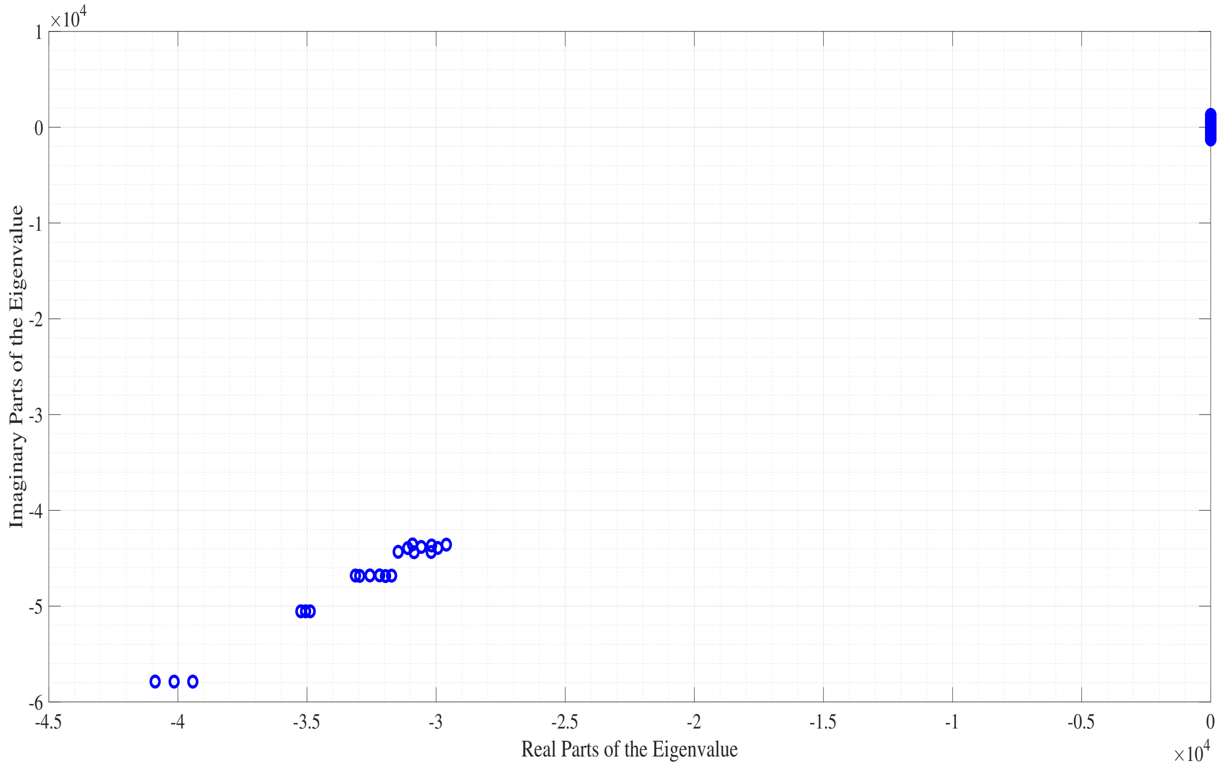

Figure 6: Eigenvalue plot of the A matrix of small signal analysis

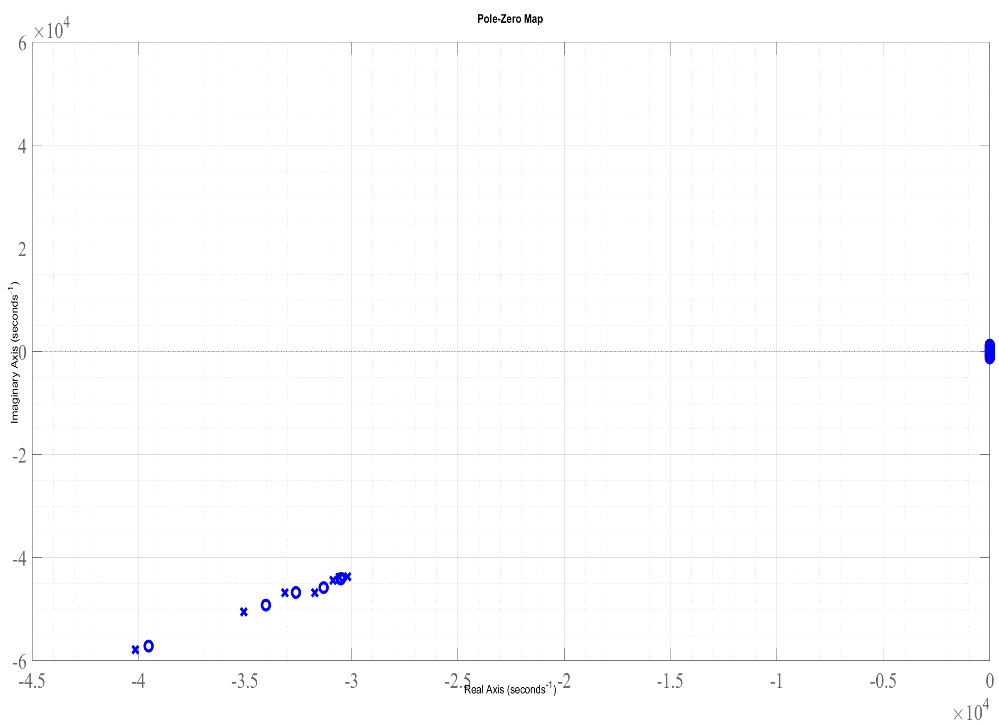

Figure 7: Pole-Zero Plot of the transfer function between $\Delta n_{ua_{-3}}$ and $\Delta v_{dc_{-3}}$

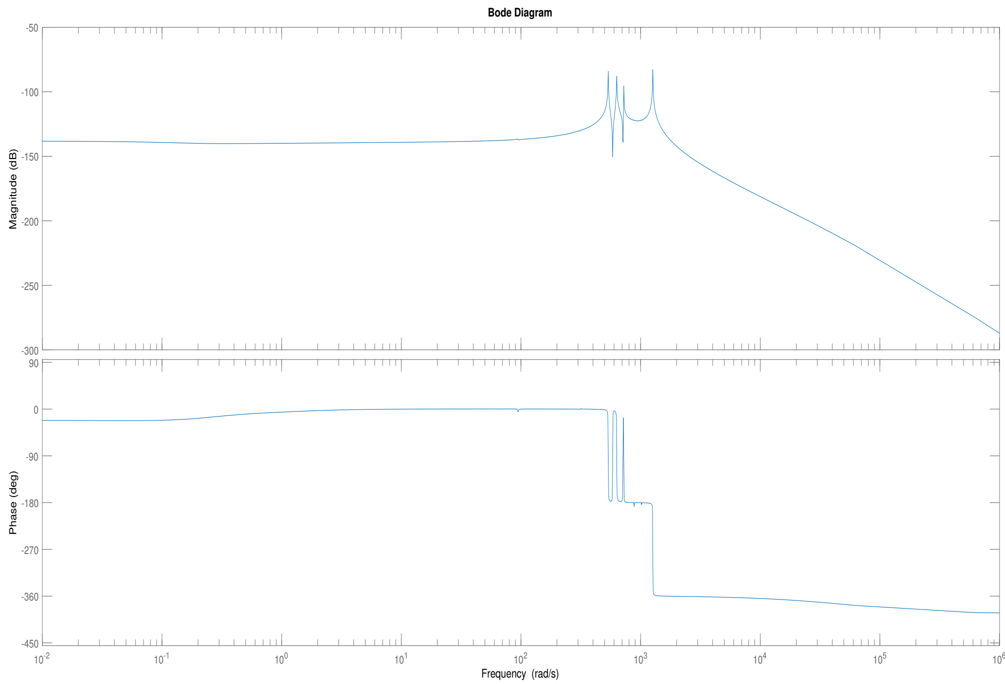

Figure 8: Bode Plot of the transfer function between $\Delta n_{ua_{-3}}$ and $\Delta v_{dc_{-3}}$

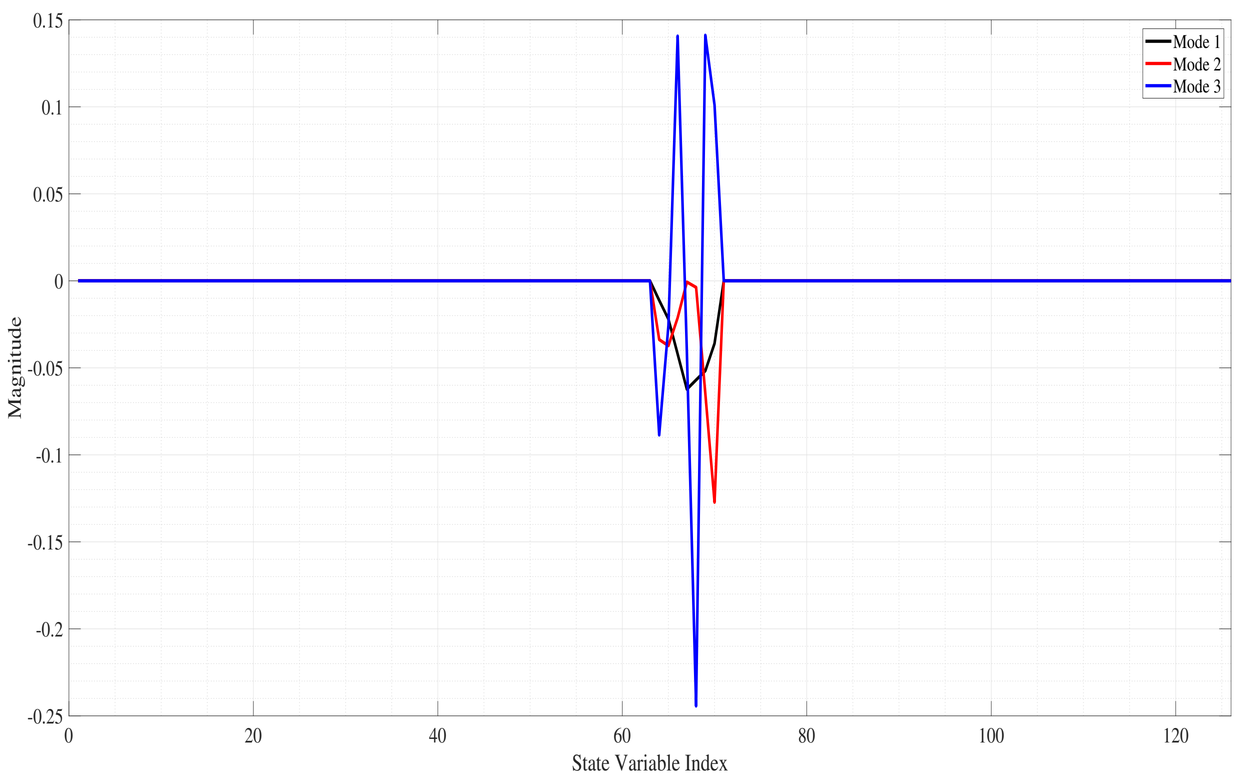

Figure 9: Participation of the first three eigenvalues on the components of $\Delta i_{ga} = [\Delta i_{ga_{-3}}, \Delta i_{ga_{-2}}, \Delta i_{ga_{-1}}, \Delta i_{ga_0}, \Delta i_{ga_1}, \Delta i_{ga_2}, \Delta i_{ga_3}]$
.